

Requirements Table

Pet Adoption & Shelter Management System — Lab 1: Requirements Engineering & UML Use-Case Modelling
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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Shelter Staff to log an animal's vaccination and medical intake records, and update the animal's availability status to “Ready for Adoption” once records are complete.	High	Pass: An animal's status changes to “Ready for Adoption” only after a complete medical record is saved. Fail: An animal under medical quarantine can be marked as adopted.	Protects animal welfare and prevents unsafe adoptions of animals still under medical hold.
FR-002	Functional	The system shall allow an Adoption Applicant to submit an adoption application for a specific animal, capturing household composition, prior pet-ownership history, and housing details.	High	Pass: A submitted application appears in the staff review queue with every mandatory field completed and linked to the correct animal. Fail: An application is accepted for review with one or more mandatory fields blank.	Structured, complete applicant data is required before staff can screen a household for suitability.
FR-003	Functional	The system shall allow Shelter Staff to review a submitted adoption application and record an Approve or Reject decision together with the reviewer's identity, a timestamp, and optional comments.	High	Pass: An approved or rejected application stores the reviewer ID, decision timestamp, and comments, and the applicant is notified within 24 hours. Fail: A decision is saved without a reviewer ID or timestamp.	Creates an auditable trail for adoption decisions and keeps applicants informed.
FR-004	Functional	The system shall allow Shelter Staff to create and track a foster home placement for an animal, recording the start date, expected end date, and foster caregiver contact details.	Medium	Pass: A foster placement record cannot be saved unless it has a start date, an expected end date, and a linked caregiver contact. Fail: A placement is created with a missing end date or no caregiver reference.	Prevents animals from being lost track of while in temporary foster care.
FR-005	Functional	The system shall allow an Adoption Applicant to search and filter the pet catalog by species, age, breed, and compatibility attributes such as “Good with Kids.”	High	Pass: Search results include only animals that satisfy every filter attribute selected by the applicant. Fail: One or more returned animals fail to match a selected filter.	Core discovery functionality that lets applicants efficiently find a suitable pet.
NFR-001	Nonfunctional	The pet search catalog must support multi-attribute filtering (Species, Age, Breed, Good with Kids) with a page response time under 200 ms.	High	Pass: Benchmark tests confirm the target latency and security standards are met under simulated peak load. Fail: p95 response time for a filtered search exceeds 200 ms.	Fast, reliable search is essential to a usable public-facing adoption catalog.
NFR-002	Nonfunctional	The system shall encrypt all applicant personal data (contact details, household information) at rest and in transit, restricting medical-record edit access to authenticated Shelter Staff accounts only.	High	Pass: A security scan confirms TLS is enforced on all endpoints and data-at-rest is encrypted; an unauthenticated request to edit a medical record is rejected with an authorization error. Fail: Any endpoint transmits data unencrypted or allows an unauthenticated edit.	Applicant and medical data are sensitive; protecting them maintains trust and regulatory compliance.

Preconditions

- The Adoption Applicant has an active account and is authenticated in the system .
- The target animal's status is “Ready for Adoption” its medical intake record has been completed and logged by Shelter Staff (see UC-04).
- The applicant does not already have a pending application for the same animal.
- The animal is not already reserved under an approved application for another applicant.

Postconditions

Success (guaranteed):

- A new adoption application record exists with status “Submitted,” a unique application ID, and a creation timestamp.
- The application is linked to the applicant's account and to the target animal, and appears in the Shelter Staff review queue (UC-03).
- The animal's listing is flagged “Application Pending” so other applicants see its status before applying.
- The applicant receives an on-screen confirmation and a confirmation email summarizing the next steps and expected review time.

Failure (minimal guarantee):

- No application record is created; the animal's status and any other applicants' pending applications are left unchanged.
- The applicant is shown the specific reason submission did not succeed and may retry.

Business Rules

- BR-1: An applicant may have at most one open (Submitted / Under Review) application per animal at a time.
- BR-2: An animal accepting new applications may hold multiple concurrent “Submitted” applications from different applicants until Shelter Staff approves one, at which point all other open applications for that animal are auto-rejected with a notification.
- BR-3: Only animals with status “Ready for Adoption” are eligible to receive applications.
- BR-4: Mandatory application fields (household composition, housing type, prior pet-ownership history, contact information) must all be completed before submission is allowed.

Main Success Scenario

1. The Adoption Applicant selects “Apply to Adopt” on the profile page of an animal marked “Ready for Adoption.”
2. The system verifies the applicant is authenticated (includes UC-07: Authenticate User); an already-authenticated session proceeds directly to step 3.
3. The system checks eligibility: the animal is still “Ready for Adoption” and the applicant has no existing open application for it (BR-1, BR-3).
4. The system displays the adoption application form, pre-filled with the applicant's saved profile details where available.
5. The applicant enters or confirms household composition, prior pet-ownership history, housing details (own/rent, yard, other pets in the home), and preferred contact information.
6. The applicant reviews the completed form and selects “Submit Application.”
7. The system validates that all mandatory fields are complete and correctly formatted (BR-4).
8. The system creates the application record with status “Submitted,” timestamps it, and links it to the applicant and the selected animal.
9. The system places the application in the Shelter Staff review queue and flags the animal's listing as “Application Pending.”
10. The system displays a confirmation message and sends a confirmation email summarizing next steps and typical review turnaround.
11. Use case ends successfully.

Alternate Flows

2a. Applicant Is Not Authenticated

- 2a1. The system redirects the applicant to the login/registration screen.
- 2a2. On successful authentication, the system returns the applicant to the animal's profile page and resumes at step 3.
- 2a3. If the applicant cancels login, the use case ends without a submission.

3a. Animal No Longer Eligible

- 3a1. The system detects the animal's status has changed since the applicant began the process.
- 3a2. The system informs the applicant the animal is no longer accepting applications and suggests similar available animals (UC-01).
- 3a3. Use case ends without a submission.

7a. Mandatory Fields Missing or Invalid

- 7a1. The system highlights each incomplete or invalid field with an inline error message.
- 7a2. The applicant corrects the flagged fields and selects “Submit Application” again, returning to step 7.
- 7a3. If the applicant abandons the form without correcting the errors, no application record is created and the use case ends without success.

8a. Duplicate Submission Detected

- 8a1. The system detects the applicant already has an open application for this animal (e.g., submitted from another browser tab).
- 8a2. The system blocks the duplicate, displays the existing application's status, and offers a link to view it.
- 8a3. Use case ends without creating a new record.

Exceptions

- System unavailable at submission: the form's entered data is retained locally and the applicant is prompted to retry; no partial record is created.

Email confirmation fails to send: the application is still saved successfully, and the confirmation is also shown on-screen and stored in the applicant's account notifications.